Hope: Transparent Microfinance Loan Management Web Application

Software Development Project Report
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Submitted By

Name	ID
Abdul Hakim Shifat	23524202129
SM Shahrier Emon	23524202095
MD Mainul Islam	23524202047
Sarowar Rony	23524202051

Introduction

Hope is a full-stack web application developed to support a transparent and organized microfinance loan management workflow. The project focuses on reducing manual dependency in borrower registration, loan application review, installment tracking, and overdue follow-up. It provides separate role-based dashboards for borrowers, field officers, supervisors, and administrators. The system was designed as an MVP for the Bangladeshi microfinance context, using a mock payment review workflow instead of real banking or payment gateway integration.

Project Idea

The submitted project idea was to build **Hope**, a transparent microfinance loan management web application for institutions that currently depend on manual loan records and field collection processes. In the proposed workflow, borrowers can register, complete their profiles, apply for predefined institutional loan products, submit installment payments, and view a clear ledger of paid, due, overdue, and remaining amounts. Field officers become involved mainly when borrowers miss payments or need offline assistance, while supervisors and admins monitor approvals, payments, overdue cases, and audit records.

The implemented project follows the main idea of the proposal and converts it into a working MVP. It includes borrower onboarding, borrower verification, predefined loan products, loan application approval, automatic repayment schedule generation, mock payment submission, payment approval, receipt creation, ledger visibility, overdue detection, case assignment, field officer visit logs, audit logs, and role-based access control. Features such as real OTP verification, NID verification through government services, real payment gateway integration, SMS notification, GPS tracking, and AI credit scoring were kept outside the MVP scope.

Features of the Project

Area	Implemented Features

Authentication and Roles	Registration, login, JWT authentication, protected routes, and role-based access for Borrower, Field Officer, Supervisor, and Admin.
Borrower Profile	Borrower profile creation with personal, occupation, income, NID, nominee, and verification information.
Borrower Verification	Supervisor/Admin can approve or reject borrower profiles with verification status tracking.
Loan Products	Predefined loan products with minimum amount, maximum amount, service charge, duration, installment frequency, late fee, eligibility note, and status.
Loan Applications	Borrowers can apply for suitable loan products; Supervisor/Admin can approve or reject applications.
Loan Generation	Approved applications create active loans and automatic installment schedules.
Installment Tracking	Installments show due dates, paid amount, outstanding amount, and status such as paid, due, and overdue.
Mock Payment Flow	Borrowers submit mock installment payments using methods such as bKash, Nagad, Rocket, Card, or Cash Assist.
Payment Review	Supervisor/Admin reviews pending payments, approves valid submissions, rejects invalid ones, and updates installment records.
Receipts	Approved payments generate receipt records that borrowers can view later.
Ledger	Borrowers can view transparent loan summaries, total payable, total paid, remaining balance, next due date, and repayment history.
Overdue Management	The system detects overdue installments and supports assignment of cases to field officers.
Field Officer Workflow	Field officers see assigned cases and submit visit logs after borrower follow-up.
Audit Logs	Important actions such as approvals, rejections, payments, and case operations are recorded for accountability.

Design of the Project

The project follows a client-server architecture with a React frontend, an Express REST API, and a MongoDB database. The frontend is organized into reusable components, protected routes, context-based authentication, and separate pages for each user role. The backend contains modular route files, middleware for authentication and error handling, Mongoose models for all core entities, and utility functions for loan calculation, overdue detection, and audit logging.

The visual design uses a responsive dashboard-based layout. Borrower screens are designed around clarity, showing loan progress, payment options, repayment schedules, and ledger information in a simple sequence. Supervisor and admin screens use tab-based operational panels for borrower verification, loan application review, pending payment review, overdue assignment, field logs, audit logs, user management, and product management. Field officer screens are kept task-focused so assigned overdue cases and visit updates can be handled quickly.

High-Level Architecture

Layer	Main Responsibility
Frontend	Presents role-based interfaces, forms, dashboards, ledger views, and API-connected workflows.
Backend API	Handles authentication, authorization, validation, business rules, loan workflows, and audit logging.
Database	Stores users, borrower profiles, loan products, applications, loans, installments, payments, receipts, cases, visit logs, and audit logs.
Utilities	Calculates loan totals and schedules, refreshes overdue installments, and records audit actions.

Main Data Entities

Entity	Purpose
User	Stores account credentials, contact information, role, and status.
BorrowerProfile	Stores borrower verification and eligibility details.
LoanProduct	Defines institutional product rules and limits.
LoanApplication	Tracks borrower loan requests and review decisions.
Loan	Stores approved loan details and total payable amount.
Installment	Stores repayment schedule, due amount, paid amount, due date, and status.
Payment	Stores mock payment submissions and approval status.
Receipt	Stores receipt details for approved payments.
OverdueCase	Tracks overdue installment follow-up assigned to field officers.
VisitLog	Records field officer borrower visits.
AuditLog	Maintains accountability for sensitive system actions.

Development of the Project

The project was implemented as a full-stack MVP with a separated frontend and backend structure. Requirements were derived from the submitted proposal and converted into role-based workflows for borrowers, field officers, supervisors, and admins. The backend was built first around Express routes, Mongoose models, authentication middleware, and business utilities for loan calculation and overdue detection. Database schemas were created for users, borrower profiles, loan products, applications, loans, installments, payments, receipts, overdue cases, visit logs, and audit logs. Seed data was added to demonstrate realistic borrowers, loan products, active loans, paid installments, pending payments, overdue installments, and assigned cases. The frontend was developed using React, Vite, Tailwind CSS, React Router, Axios, and Lucide icons. Protected routes and authentication context were added so that each role can access only its permitted pages. Borrower screens were connected to APIs for profile, loan, installment, payment, receipt, and ledger data. Supervisor and admin dashboards were connected to review queues, case assignment tools, product management, user management, and audit logs. The system was tested with seeded demo accounts and screenshots were captured from the working local application.

Development Tools, Languages, and Software

Category	Tools / Languages / Software Used
Frontend Language	JavaScript, JSX, HTML5, CSS3
Frontend Framework	React.js with Vite
Frontend Styling	Tailwind CSS
UI Icons	Lucide React
Routing	React Router DOM
API Client	Axios
Backend Language	JavaScript running on Node.js
Backend Framework	Express.js
Database	MongoDB
ODM	Mongoose
Authentication	JSON Web Token and bcryptjs
Backend Security / Middleware	Helmet, CORS, Morgan, custom auth middleware
Development Environment	Visual Studio Code, npm, local terminal
Testing / Demo Support	Seed script, local MongoDB, browser testing, Playwright screenshots

Overview of the Project

This section presents selected screenshots from the developed project. The screenshots show the public landing page, borrower dashboard, borrower ledger, admin loan application review, and overdue case management workflow.

Public Home Page

Hope public home page

The home page introduces the platform and communicates the main value of transparent, role-based microfinance management.

Borrower Dashboard

Borrower dashboard

The borrower dashboard shows the borrower's profile status, repayment progress, loan summary, active loan information, payment submission area, and repayment schedule.

Borrower Ledger

Borrower ledger

The ledger screen gives borrowers a transparent view of payment progress, total payable amount, total paid amount, remaining balance, and installment-level records.

Admin Loan Application Review

Admin loan application review

The admin and supervisor workflow includes review queues for loan applications, allowing authorized users to approve or reject applications based on borrower and product information.

Overdue Case Management

Overdue case management

The overdue management screen supports assignment of overdue installments to field officers and helps the institution track follow-up actions.

Contribution

Member Name	ID	Contribution
Abdul Hakim Shifat	23524202129	Led project planning and requirement mapping from the proposal; designed the backend workflow for authentication, loan applications, loan approval, repayment schedules, and API structure; coordinated final integration and report preparation.
SM Shahrier Emon	23524202095	Developed and refined frontend pages for public navigation, borrower dashboard, login/register flow, loan product browsing, payment submission, ledger view, and responsive interface behavior.
MD Mainul Islam	23524202047	Worked on database design and backend data modeling for users, borrower profiles, loan products, loans, installments, payments, receipts, overdue cases, visit logs, and audit logs; supported seed data preparation for demonstrations.
Sarowar Rony	23524202051	Contributed UI/UX review, dashboard usability improvements, role-based workflow testing, screenshot preparation, bug reporting, and validation of borrower, supervisor/admin, and field officer use cases.

Conclusion

Hope successfully demonstrates how a microfinance institution can digitize loan application, approval, repayment monitoring, ledger transparency, and overdue follow-up within a single web platform. The completed MVP preserves the central idea of the original proposal while keeping high-risk external integrations, such as real OTP, NID verification, and payment gateway services, outside the project scope. The role-based design improves clarity for borrowers and gives institutional users better control over approvals, cases, products, users, and audit records. Future development can extend the system with real payment gateway integration, SMS/OTP services, document upload, deployment, analytics, and stronger production-grade security controls.